

Assembly Guide

Localization of a Mobile Robot

Author: Srikanth Nanda
Course of Studies: Business Intelligence and Data Analytics

First examiner: Prof. Dr. Elmar Wings

Second examiner: Prof. Dr. Jonas Schwarz

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Acronyms

1 Hints

1.1 Operating Best Practices

- Initialize the localization system in a well-lit environment for optimal visual feature detection
- Allow calibration time (typically 5-10 seconds) before trusting position estimates
- Keep the environment consistent during long test runs to minimize loop closure challenges
- Monitor battery voltage; localization accuracy degrades with insufficient power
- Record baseline trajectories for ground truth comparison
- Document environmental conditions (lighting, texture, scale features) for experiment reproducibility

1.2 Localization Reliability

- High-texture environments yield better feature matching
- Large open spaces require more robust loop closure detection
- Rapid rotations may cause temporary tracking loss
- Repeated patterns can confuse feature matching; mark unique landmarks

2 Main Function

2.1 Introduction to the Localization System

The mobile robot localization system enables autonomous navigation by continuously estimating the robot's position and orientation within its environment. This manual covers operating the robot, executing localization algorithms, and interpreting results.

2.2 Core Operating Modes

1. **Data Collection**: Record environmental images and sensor data for offline processing
2. **Real-time Localization**: Continuous position estimation during navigation
3. **Mapping Mode**: Build environmental maps for reference-based localization
4. **Testing Mode**: Validate localization accuracy against known paths

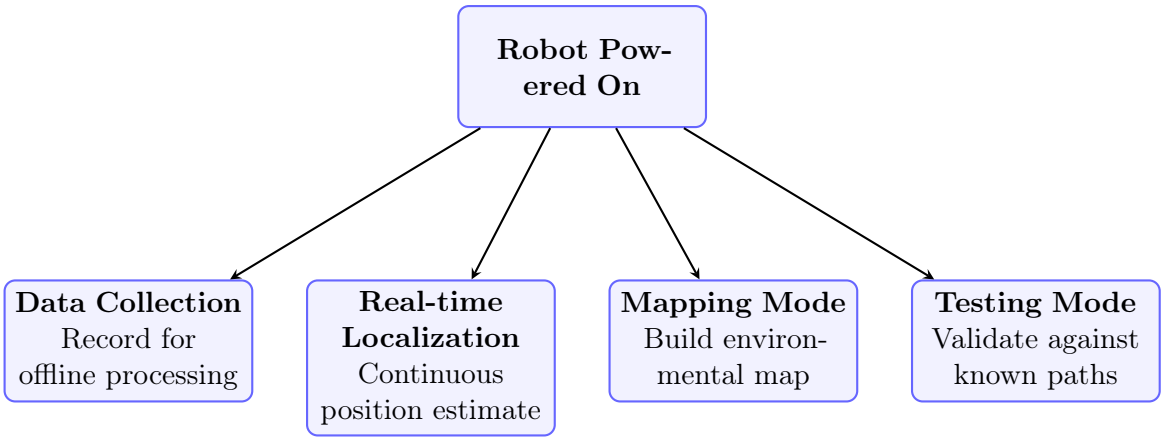


Figure 2.1: Four core operating modes available after robot power-on

2.3 Typical Workflow

Start with environmental setup, initialize the localization system, execute the desired operating mode, and monitor sensor outputs through the provided interface.

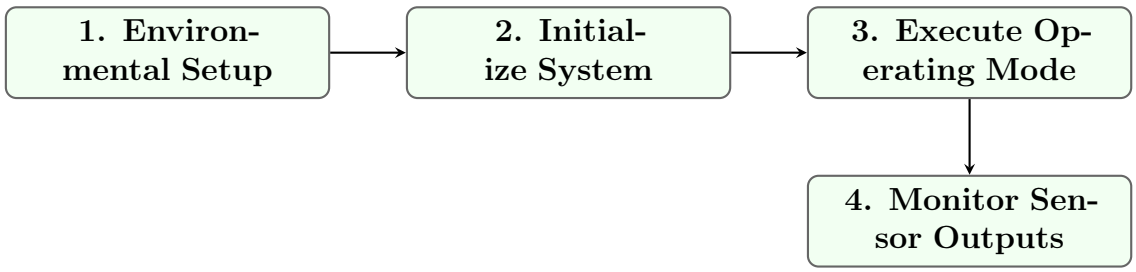
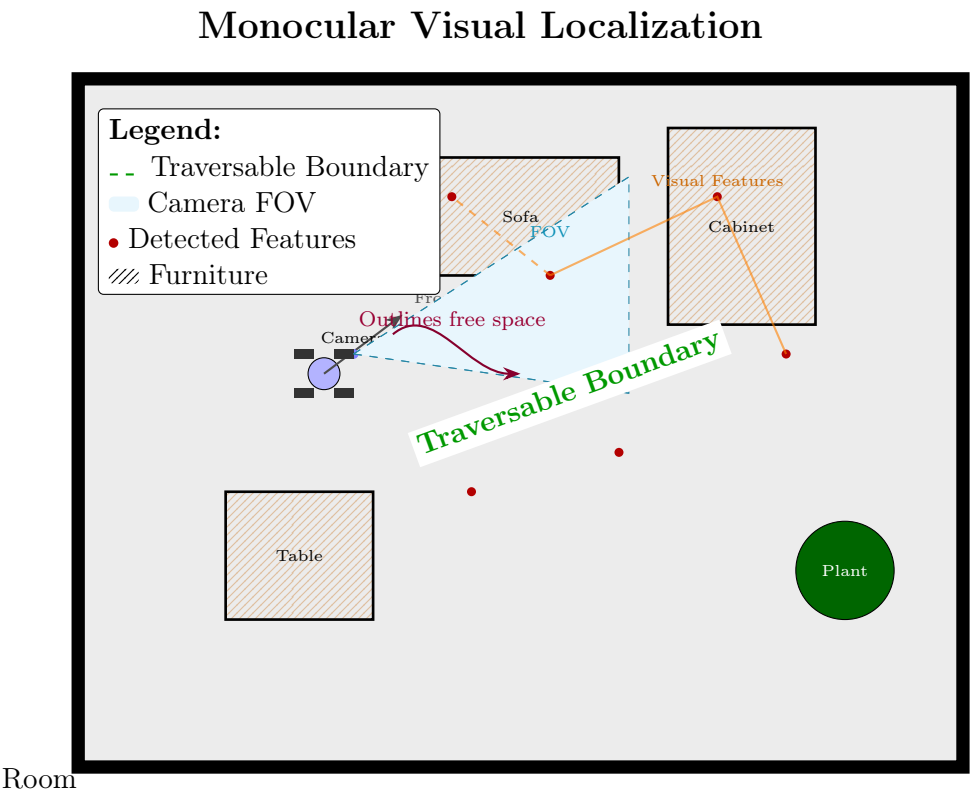
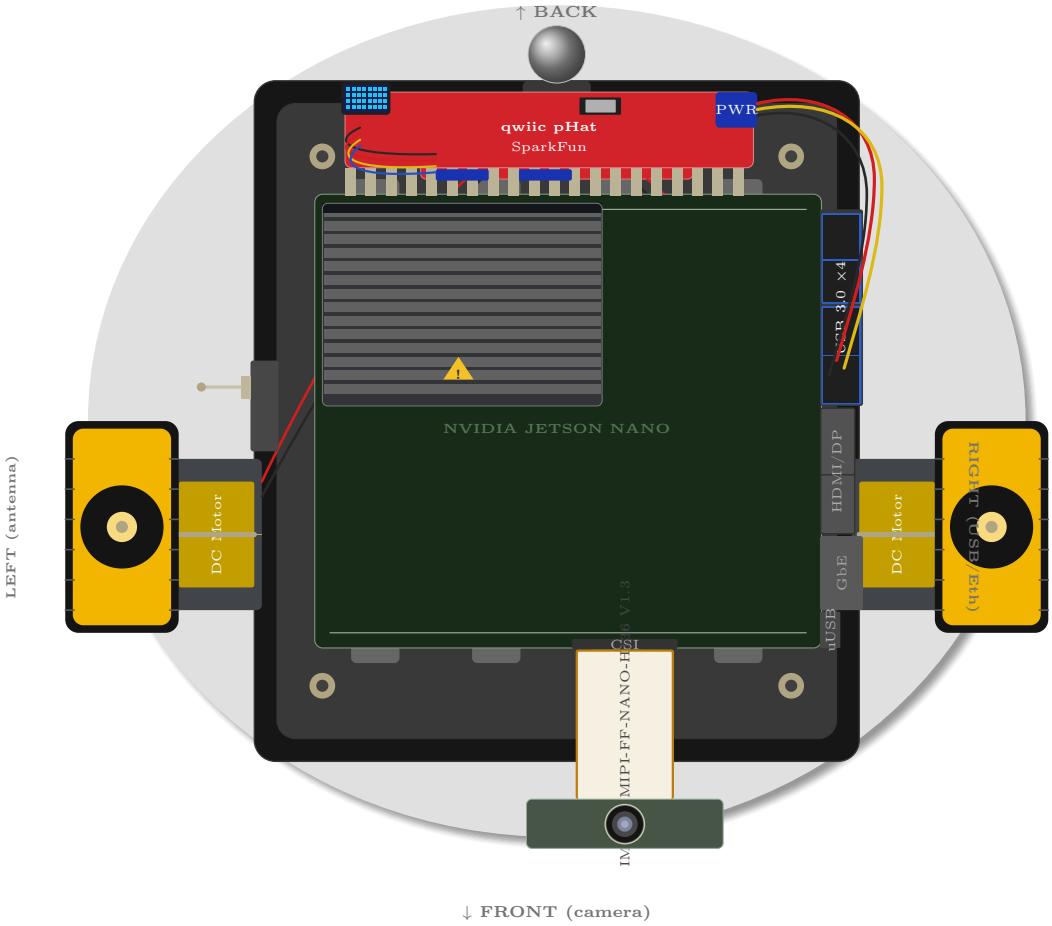


Figure 2.2: Typical operating workflow from environmental setup to sensor monitoring

3 Sketch


Monocular Visual Localization
Legend:
Traversable Boundary
Camera FOV
Detected Features
Furniture
Sofa
FOV
Visual Features
Cabinet
Outlines free space
Traversable Boundary
Table
Plant
Room



SparkFun JetBot AI Kit — Top View

4 List Of Parts

4.1 Software Components

- **Vision Pipeline**: OpenCV with SIFT/ORB feature extraction
- **Odometry Engine**: Visual and encoder-based odometry
- **SLAM Framework**: g2o-based pose graph optimization
- **Mapping Module**: Occupancy grid or feature map representation
- **User Interface**: Web dashboard for real-time monitoring

4.2 Tools and Utilities

- Ground truth measurement device (optical tape measure or known markers)
- Test environment with distinctive visual features
- Calibration checkerboard pattern
- Trajectory logging and analysis tools

5 Set Up

5.1 System Initialization

1. Power on the robot and wait for Jetson Nano to boot (30-60 seconds)
2. Connect to robot via SSH: `ssh jetson@<robot_ip>`
3. Launch localization software: `roslaunch localization system.launch`
4. Open web dashboard: `http://<robot_ip>:8080`

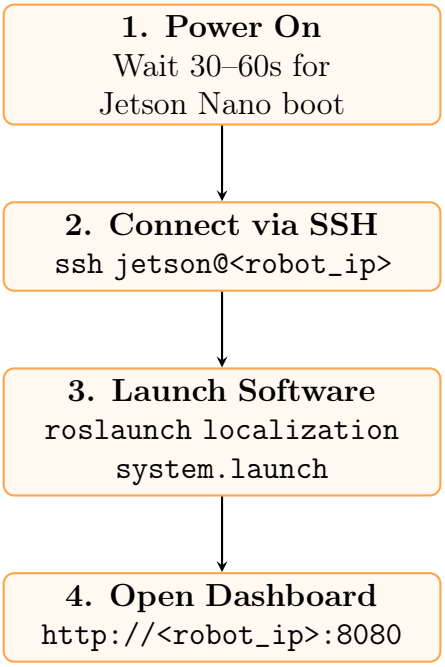


Figure 5.1: System initialization sequence from power-on to web dashboard access

5.2 Environment Preparation

- Place robot in test arena with known dimensions
- Ensure adequate lighting (minimum 500 lux recommended)
- Add visual landmarks or textured features

- Clear obstacles if in autonomous mode

5.3 Calibration Procedure

1. Run camera calibration checkerboard scan
2. Perform encoder zero calibration
3. Execute test path to verify baseline accuracy
4. Adjust feature extraction parameters if needed

6 Functions

6.1 Core Localization Features

6.1.1 Visual Odometry Mode

- Estimates robot motion from consecutive camera frames
- Output: Relative pose transformation (translation and rotation)
- Good for short-term trajectory (< 20 meters)

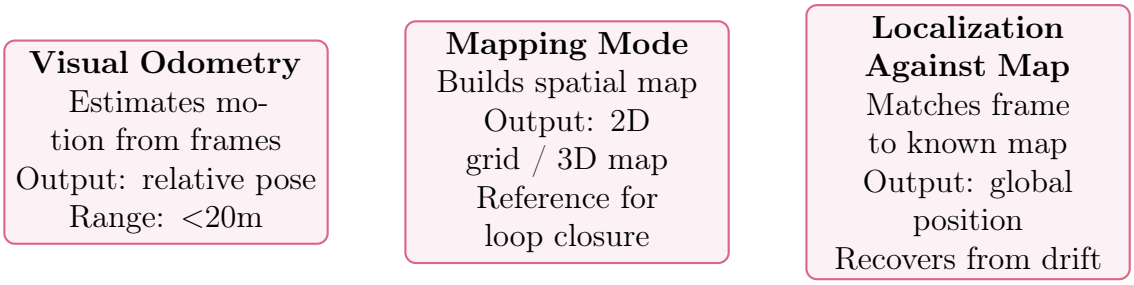


Figure 6.1: Comparison of the three core localization features and their outputs

6.1.2 Mapping Mode

- Builds spatial map of environment from visual features
- Creates reference database for loop closure detection
- Output: 2D occupancy grid or 3D feature map

6.1.3 Localization Against Map

- Matches current camera frame against known map
- Estimates global position even without encoder data
- Enables recovery from encoder drift

6.2 Output Data

- Current position (x, y, theta) in world coordinates
- Confidence estimate for position accuracy
- Trajectory history (points visited)
- Loop closure events (when revisiting areas)

7 Maintenance

7.1 Localization System Validation

7.1.1 Periodic Recalibration

- Recalibrate camera intrinsics monthly or after impacts
- Verify encoder tick-to-distance conversion on test path
- Rerun visual odometry on known trajectory to check drift

7.1.2 Accuracy Assessment

1. Operate robot along marked reference path
2. Compare estimated trajectory against ground truth
3. Calculate Absolute Trajectory Error (ATE)
4. If ATE exceeds threshold, investigate sensor issues

7.2 Feature Database Maintenance

- Clear cached feature maps monthly
- Rebuild features if environment significantly changes
- Archive training trajectories for future reference

7.3 Log Monitoring

- Monitor for repeated feature extraction failures
- Track loop closure detection frequency
- Check for encoder dropout events

8 Safety Guidelines

8.1 Autonomous Operation Safety

- Always supervise autonomous runs; be prepared to stop the robot manually
- Establish safe boundaries and clear obstacles from test area
- Never combine autonomous mode with foot traffic in same area
- Use speed limiter on first runs in unfamiliar environments

8.2 Testing Precautions

- Start with low motor speeds and gradually increase
- Test collision avoidance in controlled environment before deployment
- Do not rely solely on autonomous features; maintain manual override capability
- Report any unsafe behaviors immediately and disable autonomous mode

8.3 Environmental Hazards

- Avoid stairs, ledges, and elevated surfaces unless tested
- Keep away from water sources
- Test on surfaces similar to intended deployment (carpet, tile, concrete)
- Be aware of differences between testing and real environments

8.4 Emergency Procedures

- If robot malfunctions during testing, disconnect power immediately
- Document any near-miss incidents for analysis
- Do not attempt repairs while system is powered

9 Troubleshooting

9.1 Localization Accuracy Issues

- **High position drift**: Increase feature extraction quality or optimize camera calibration

- **Loop closure fails**: Environment lacks distinctive features; add markers or increase detection sensitivity

- **Encoder drift dominates**: Visual features may be unreliable; check lighting or add reference landmarks

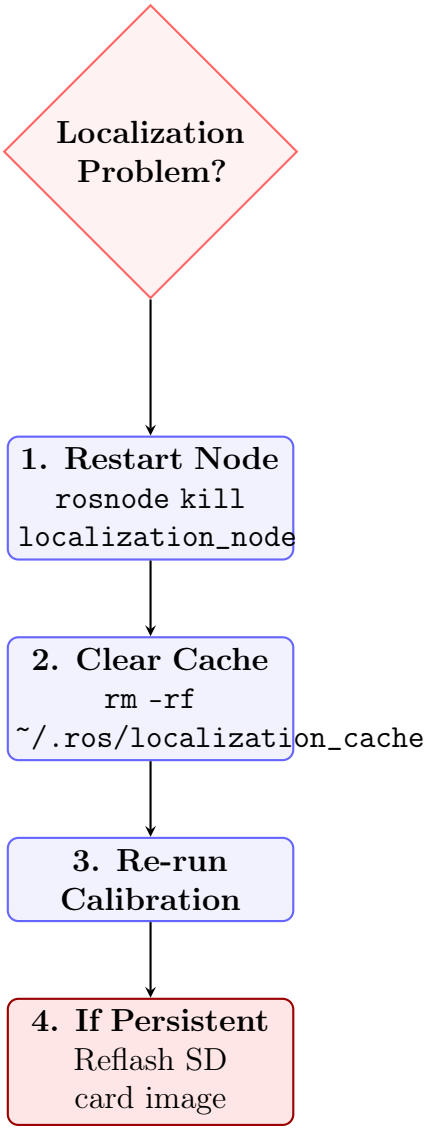


Figure 9.1: Recovery workflow for resolving persistent localization problems

9.2 Performance Problems

- **Slow frame processing**: Too many features extracted; adjust threshold
- **High CPU usage**: Disable feature visualization or reduce resolution
- **Memory exhaustion**: Prune old feature maps from memory; consider streaming to disk

9.3 Sensor Failures

- **Encoder dropout**: Check wheel spin under no load; verify encoder connections
- **Feature extraction fails**: Inadequate lighting; switch to robust feature detector (ORB vs SIFT)
- **System crashes**: Log dump file location provided in error message; report to development team

9.4 Recovery Steps

1. Restart localization system: `rosnode kill localization_node`
2. Clear cached data: `rm -rf  /.ros/localization_cache`
3. Re-run calibration procedure
4. If persistent, reflash SD card image

10 Add Ons

10.1 Enhanced Localization Features

10.1.1 Multi-Sensor Fusion Mode

- When IMU available: Uses gyro for drift-free heading
- When encoders available: Fuses with visual odometry for robustness
- When both available: Achieves best-in-class accuracy
- Enable via config: `sensor_fusion: true`

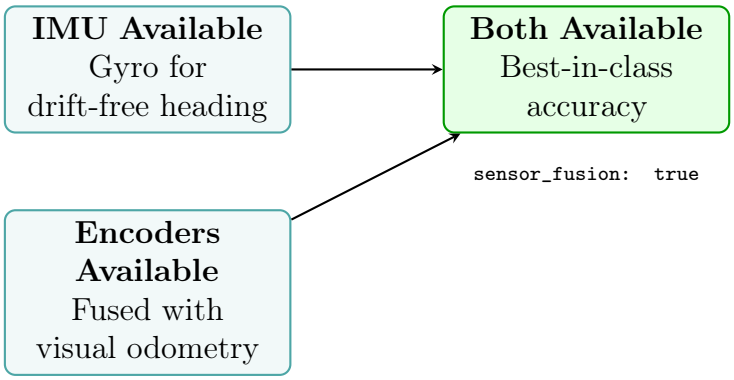


Figure 10.1: Multi-sensor fusion logic combining IMU and encoder data for improved accuracy

10.1.2 3D Mapping and Height Awareness

- Stereo camera provides depth for each visual feature
- Enables stair and ramp detection
- Improves obstacle avoidance in complex 3D environments
- Requires `mapping_3d: true` in config

10.1.3 Global Localization (Lidar-based)

- Use Lidar scans to match against precomputed map occupancy grids
- Enables position recovery if visual tracking is lost
- Can relocalize robot anywhere in mapped environment
- Access via web dashboard: click G̈lobal Relocalizeb̈utton

10.2 Performance Tuning Options

- Reduced feature extraction rate for low-power mode (15 Hz instead of 30 Hz)
- Aggressive loop closure for tightening accuracy `loop_closure_aggressiveness: high`
- Conservative mode for safety (requires strong evidence before transitions)